VIJAYABHERI

Malappuram District Panchayat Educational Project



LEAP-26

ENGLISH MEDIUM

PHYSICS

SSLC

QUESTION POOL

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MALAPPURAM DISTRICT PANCHAYATH
EDUCATIONAL PROJECT

LEAP 26

SSLC PHYSICS QUESTION POOL

അഭിമുഖം

2001–2002 വർഷത്തിൽ മലപ്പുറം ജില്ലയുടെ സമഗ്ര വിദ്യാഭ്യാസ പുരോഗതി ലക്ഷ്യം വെച്ചു മലപ്പുറം ജില്ലാ പഞ്ചായത്ത് ആസൂത്രണം ചെയ്തു നടപ്പിലാക്കിവരുന്ന പദ്ധതിയാണ് വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതി.
കഴിഞ്ഞ 24 വർഷത്തിനുള്ളിൽ ജില്ലയുടെ വിദ്യാഭ്യാസ മുഖച്ഛായ മാറ്റാൻ ഈ പദ്ധതിയിലൂടെ കഴിഞ്ഞിട്ടുണ്ട്. എസ്എസ്എൽസി വിജയശതമാനത്തിലും പ്ലസ് ടു വിജയശതമാനത്തിലും മത്സര പരീക്ഷകളിലും ഏറെ പിന്നിലായിരുന്ന ജില്ല ഇന്ന് സംസ്ഥാനത്തെ മറ്റേത് ജില്ലയോടും കിടപിടിക്കാവുന്ന രീതിയിൽ മാത്രമല്ല, പല മേഖലയിലും അവരെക്കാൾ മുന്നേറാൻ ഇന്ന് ജില്ലയ്ക്ക് കഴിഞ്ഞിട്ടുണ്ട്.
ഏറ്റവും കൂടുതൽ എ പ്ലസ്സുകളുള്ള ജില്ല, ഏറ്റവും കൂടുതൽ കുട്ടികൾ കേന്ദ്ര സർവകലാശാലയിലേക്ക് പ്രവേശനം നേടുന്ന ജില്ല, മെഡിക്കൽ എൻജിനീയറിങ് എൻട്രൻസ് പരീക്ഷകളിൽ മുന്നിൽ നിൽക്കുന്ന ജില്ല ഇന്ന് മലപ്പുറം ജില്ലയാണ്. വിദ്യാഭ്യാസപരമായി പിന്നാക്കം നിൽക്കുന്ന ജില്ല എന്ന അപഖ്യാതി നമ്മളിന് മറികടന്നു.

മറ്റ ജില്ലകളിൽ നിന്ന് ഭിന്നമായി ജില്ലയിലെ എല്ലാ തദ്ദേശ സ്വയംഭരണ സ്ഥാപനങ്ങൾക്കും എംഎൽഎമാർക്കും ഇന്ന് പ്രത്യേക വിദ്യാഭ്യാസ പദ്ധതികളുണ്ട്. ഇത്തരം എല്ലാ പദ്ധതികൾക്കും പ്രചോദനമായത് വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതിയാണ്. സംസ്ഥാനതലത്തിൽ മാത്രമല്ല ദേശീയ തലത്തിൽ പോലും ഈ പദ്ധതി ചർച്ച ചെയ്യപ്പെട്ട വരുന്നു. യൂണിവേഴ്സിറ്റികളിൽ വിജയിഭേരിപദ്ധതി ഗവേഷണ വിദ്യാർത്ഥികൾക്ക് പഠന വിഷയമാണ്.

ഈ പദ്ധതിയുടെ വിജയം മലപ്പുറം ജില്ലാ പഞ്ചായത്തിന്റെ വിജയമല്ല, ഇത് ഏറ്റെടുത്ത അധ്യാപകരുടെയും രക്ഷിതാക്കളുടെയും വിദ്യാർത്ഥികളുടെയും പിന്തുണയാണ് ഈ പദ്ധതിയുടെ വിജയത്തിന്റെ അടിസ്ഥാനം. അതുകൊണ്ടുതന്നെ അവരോടുള്ള കടപ്പാട് ഈ അവസരത്തിൽ പ്രത്യേകം രേഖപ്പെടുത്തുന്നു.

കഴിഞ്ഞ വർഷങ്ങളിലേതുപോലെ ഈ വർഷം നിലവിൽ വന്ന പുതിയ ജില്ലാ പഞ്ചായത്ത് ബോർഡും വിജയഭേരി പദ്ധതി കൂടുതൽ ഊർജ്ജസ്വലമായി മുന്നോട്ടുകൊണ്ടുപോകാൻ തീരുമാനിച്ചിട്ടുണ്ട്. അതിന്റെ ആദ്യപടിയായി ഈ വർഷം എസ്എസ്എൽസി പരീക്ഷ എഴുതുന്ന വിദ്യാർത്ഥികൾക്കും അധ്യാപകർക്കും സഹായകരമായ രീതിയിൽ എല്ലാ വിഷയങ്ങളുടെയും ചോദ്യശേഖരം പുറത്തിറക്കുകയാണ്.

പുതിയ പാഠപുസ്തകങ്ങളും പുതിയ രീതിയിലുള്ള പരീക്ഷ രീതിയുമാണ് ഈ വർഷത്തെ വിദ്യാർഥികൾ അഭിമുഖീകരിക്കേണ്ടത്. അവർക്ക് ഏറെ സഹായകമായ രീതിയിൽ റിവിഷൻ നടത്തുന്നതിനും പുതിയ രീതിയിലുള്ള ചോദ്യങ്ങൾ പരിചയപ്പെടുത്തുന്നതിനുമാണ് ഇത്തരമൊരു ചോദ്യശേഖരം തയ്യാറാക്കിയിരിക്കുന്നത്. എല്ലാ വിഷയങ്ങളുടെയും എല്ലാ പാഠങ്ങളിലുമുള്ള പ്രതീക്ഷിത ചോദ്യങ്ങൾ ഇതിൽ ലഭ്യമാക്കിയിട്ടുണ്ട്. ഈ ചോദ്യ ശേഖരം വിദ്യാർത്ഥികൾക്കും അധ്യാപകർക്കും ഏറെ സഹായകരമായിരിക്കുമെന്ന് പ്രതീക്ഷിക്കുന്നു.

ദൈനംദിന പ്രവർത്തനങ്ങൾക്കിടയിൽ ഇത്തരമൊരു ഭാരിച്ച ജോലി ഏറ്റെടുത്ത് കൃത്യസമയത്ത് പൂർത്തീകരിച്ച് പ്രസിദ്ധീകരിക്കുന്നതിന് രാപ്പകലോടൊപ്പം നിന്ന അധ്യാപകരെ പ്രത്യേകം അഭിനന്ദിക്കുന്നു.

പി എ ജബ്ബാർ ഹാജി
പ്രസിഡന്റ്
മലപ്പുറം ജില്ലാ പഞ്ചായത്ത്

അഡ്വ. എ.പി. സുജിനി
വൈസ് പ്രസിഡൻറ്

പി കെ അസ്സു
ആരോഗ്യ-വിദ്യാഭ്യാസ സ്റ്റാൻഡിംഗ് കമ്മിറ്റി ചെയർമാൻ

ടി സലിം
കോഡിനേറ്റർ
വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതി

ശിൽപശാലയിൽ പങ്കെടുത്തവർ

DEEPAK C (GBHSS - MALAPPURAM)

ARUN S NAIR (CHMSS ADAKKAKUNDU)

BINDUMOL N M (SNMHSS PARAPPANANGADI)

ISHAQ MULANHIPULAKKAL (GRHSS KOTTAKKAL)

BIBIN PHILIP (MTHSS CHUNGATHARA)

SURESH K (GMVHSS NILAMBUR)

AMPILI G (VVMHSS MARAKKARA)

SABIN SEBASTIAN (VHM HSS MORAYUR)

SIMIL RAHMAN K (KMHSS KUTTOOR NORTH)

1.SOUND WAVES

1 MARK

1) Which factor affects the natural frequency?
a) Length, b) Elasticity c) Color d) Material
Which of the following is the correct statement?
i) a,c only ii) b,d only iii) a,b,d only iv) b,c,d only

2) Read the statements and write the correct conclusion?
i) Particles in longitudinal waves vibrate parallel to the direction of propagation of the wave.
ii) Transverse waves produce crests and troughs.
a) Both the statements are wrong.
b) Statement i is wrong and ii is right.
c) Both the statements are right.
d) Statement ii is right and i is wrong.

3) Which of the following frequencies of sound can a human hear?
a) 10Hz b) 17Hz c) 20Hz d) 12Hz

4) Find and fill in the relationship?
Wavelength : Meter
Frequency : --------

5) Which wave requires a medium to travel?
(Radio wave, light wave, sound wave, X-ray)

2 MARK

6) Correct and write if there is any mistake.
a) Frequency is the time required for one oscillation.

b) Frequency decreases as the length of the pendulum increases.

7) A simple pendulum completes 30 oscillations in 60 seconds. If so, calculate its period.

8) A tuned tuning fork is placed on the surface of a table. What difference does the sound make? What is the reason?

9) If a wave travels 700m in 2 seconds, calculate the wave speed?

10) At what minimum distance should the reflecting surface be to hear an echo?

MARK 3

11) Match the following.

A	B
1) Sonar	i) Resonance
2) Resonance	ii) Ultrasonic wave
3) Earthquakes	Iii) Longitudinal wave
	iv) Seismic waves

A) 1 – ii, 2 – i, 3 – iv B) 1 – iv, 2 – i, 3 – iii
C) 1 – iii, 2 – iv, 3 – i D) 1 – i, 2 – iii, 3 – ii

12) Correct if wrong.

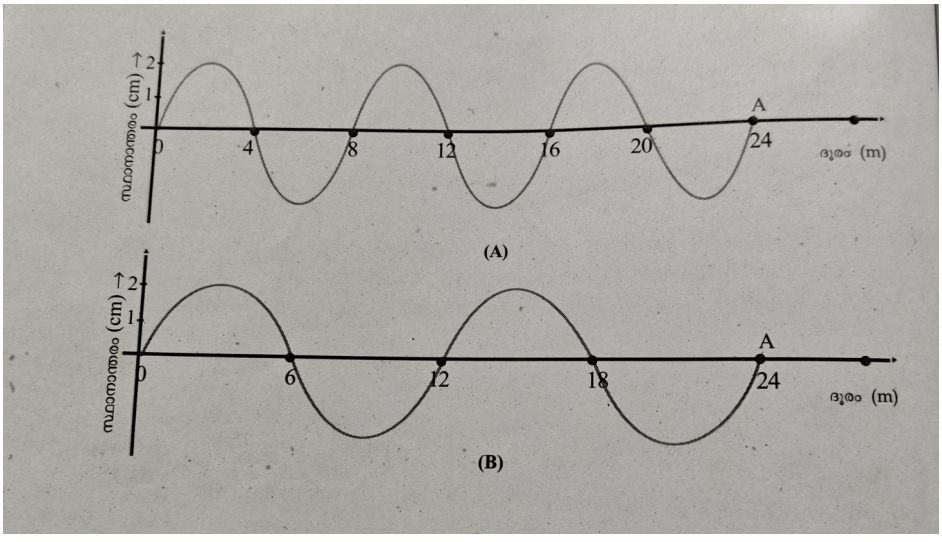
a) Ultrasonic waves are sound waves with a frequency below 20Hz.
b) When the speed is constant, the frequency of the wave is directly proportional to the wavelength.
c) Sound is a transverse wave.

13) Write three cases where resonance is used?
14) If the speed of sound in water is considered to be 1500m/s, can the echo of sound emitted to a depth of 50m be heard?

15) Find the reason.
a) Ceilings in auditoriums are made by bending.
b) Humans cannot hear the sound waves made by bats.
c) When there is a thunderstorm, the windows of buildings vibrate and make a sound.

4 MARK

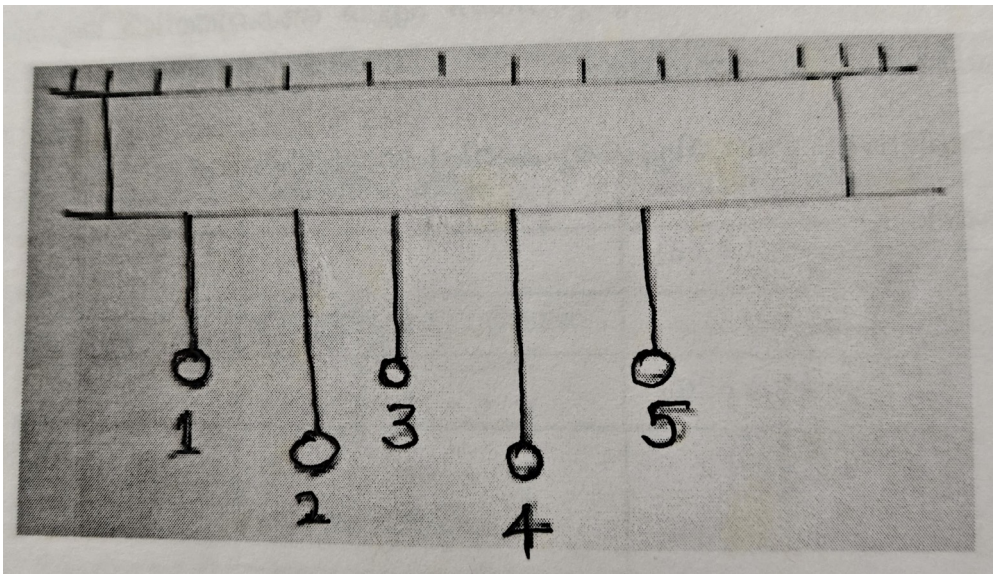
16.



Observe the picture.

If the waves travel this distance in two seconds,
a) Write the wavelength of each wave.
b) Which wave has the higher frequency?
c) Write the lengths of waves A and B.
d) Write the equation that shows the relationship between wavelength and frequency?

17.



Four bobs are suspended as shown in the figure. The first bob is made to oscillate.
a) Which bobs oscillate at the same frequency as the first bob?
b) Explain the reason why some bobs vibrate at the same frequency?
c) Soldiers are not allowed to march across a suspension bridge. What is the
reason?

18) The frequency of the sound emitted by an object is 30000Hz.
a) By what name are these sound waves known?

b) Can humans hear this sound?
c) Write two uses of this sound wave?

19) Observe the picture.



a) What do C and R stand for?
b) What type of wave is this?
c) Write two examples of such waves?

20) When speaking from inside a newly constructed hall, the sound is not heard clearly and is perceived as a hum.
a) Write the reason for this hearing experience?
b) Write a way that can be done inside the hall to avoid this hum?
c) Write two situations where the frequency response of sound is utilized?

2.LENSES

1. MARKS

1.Which type of lens is thicker at the middle?

2.What is the power of a lens with focal length 1 meter?

3.Express magnification in terms of object and image heights.

4.Which lens can diverge light rays?

5.Which lens is used as a magnifying glass?

2 MARKS

6.Explain the term 'power of accommodation' of the eye.

7.What is the difference between a real image and a virtual image?

8.What happens to the focal length when a convex lens is immersed in water? Why?

9.Define the least distance of distinct vision.

10.A student cannot see clearly objects beyond 3 meters. What type of lens should be used to correct this defect?

11.An object is placed at 2F of a convex lens. What are the characteristics of the image formed?

12.Why does the Sun appear as if it has no depth when viewed through a telescope?

13.A lens has power +5 D. Is it convex or concave? What is its focal length?

3. MARKS

14.An object is placed between the optical centre and principal focus of a convex lens. Draw the ray diagram and describe the image formed.

15.Differentiate between myopia and hypermetropia. How are they corrected?

16.What is presbyopia? How is it different from other defects of vision?

17.Explain why a real image can be projected on a screen but a virtual image cannot.

18.An object 4 cm high is placed at a distance of 15 cm from a convex lens of focal length 10 cm. Find the position, nature, and size of the image formed.

19.The power of a lens is +4D. An object is placed 30 cm from this lens. Calculate the position of the image and magnification.

20.A convex lens produces a real image which is twice the size of the object placed at 15 cm from it. Calculate: (a) focal length, (b) image distance.

4. MARKS

21.Draw a labeled diagram of a compound microscope. Explain its working with ray diagram showing image formation by objective and eyepiece. Mention the characteristics of images formed at each stage.

22.Describe the construction of a refracting telescope with a neat labeled diagram. Explain how it forms the image of a distant object. Why is the objective lens of a telescope made with larger aperture?

23 A.Can a single lens produce an image that is both magnified and inverted? Justify with ray diagram and calculations.

23.B.A student claims that virtual images cannot be photographed. Is this always true? Explain with examples.

24.**Assertion & Reasons** are given below. Say True or False

A.**Assertion:** A convex lens can form both real and virtual images. **Reason**: Real images are formed when object is beyond focus, virtual when between focus and lens.

B. **Assertion**: Power of convex lens is positive. **Reason**: Convex lens has real focus.

C. **Assertion:** Concave lens always forms virtual image. **Reason**: Concave lens diverges light rays.

D. **Assertion:** Myopia can be corrected using concave lens. **Reason**: Concave lens diverges light rays before they enter eye.

E. **Assertion**: Magnification can be greater than, equal to, or less than one. **Reason:** Magnification depends on position of object relative to lens.

3.Vision and world of colours

1.The reason the Sun appears reddish during sunrise and sunset is that red light has:
a) More scattering
b) Less scattering
c) Less wavelength
d) More energy

2.The reason for the twinkling of stars is:
a) Light pollution
b) Scattering of light
c) Atmospheric refraction
d) Total internal reflection of light .

3.The Tyndall effect is commonly observed in which type of medium?
a) Pure liquids
b) Colloidal solutions
c) Prisms
d) Flat surfaces

4.The lens used to correct Myopia (nearsightedness)is
a) Convex lens
b) Concave lens
c) Cylindrical lens
d) Bi-focal lens

5.In an eye with Hypermetropia (farsightedness), the image is formed:
a) In front of the retina
b) Behind the retina
c) On the eye lens
d) On the retina

6.The vision defect caused by the loss of power of accommodation with increasing age is:
a) Myopia
b) Hypermetropia
c) Astigmatism
d) Presbyopia

7.How many times does internal reflection occur inside a raindrop for the formation of a rainbow?
a) 1
b) 2
c) 3
d) 4

8.If Red and Cyan combine to give white light, what is the complementary colour of Red?
a) Yellow
b) Cyan
c) Magenta
d) Blue

9.The reason for the colour of an opaque object is:
a) The object absorbs all colours
b) The object transmits all colours
c) The object reflects only its own colour
d) The object reflects all colours

10.A person who needs a lens with a power of
P = -2.5D is suffering from which vision defect?
a) Hypermetropia
b) Myopia
c) Presbyopia
d) Colour Blindness

11.The reason for using red light in signals is:
a) Red light has higher speed
b) Red light has lower wavelength
c) Red light has less scattering
d) Red light has more energy .

12.Which of the following colour pairs, consisting of a primary colour and a secondary colour, combine to give white light?
a) Green, Blue
b) Blue, Yellow
c) Yellow, Red
d) Red, Cyan
.Choose the correct option from the following:
i) a, b
ii) b, d
iii) c, d

iv) b

13.Identify the pair of primary colors from the following: a) Yellow, Magenta
b) Red, Green
c) Cyan, Blue
d) Magenta, Red

. **2 marks**

14.Why does a red rose appear black in green light?

15.Why does the Sun appear red during sunrise and sunset?

16.Visible light is made up of electromagnetic waves. a) Which color has the longest wavelength in visible light? b) Which color undergoes the maximum scattering? c) What is the relationship between scattering and wavelength?

17.Why do transparent objects appear colorless?

18.What is Presbyopia? What is its remedy?

19.Why is yellow light used in fog lamps?

20.Arrange the following electromagnetic waves in increasing order of their wavelength: X-rays, Radio waves, Visible light, UV rays
a. In this which rays causes the heating effects ..
21.What change occurs to the focal length of the eye lens when looking at a distant object?

22.Yellow is a secondary color: a) Which primary colors combine to form yellow? b) What happens if three primary colors are mixed in equal proportions?

23.Write two consequences of light pollution?

24.The red of the setting sun and the beautiful rainbow make nature lovely. a) Which light phenomenon causes a rainbow? b) Which colour is seen on the inner edge of the rainbow?

25.Analyze the following statements and state the reason: a) The sky appears dark on the Moon. b) The sky in cities mostly appears grayish.

3marks

26.A person is unable to clearly see objects within one meter.

a) Identify the eye defect.
b) Write two reasons for this defect.

27.The sky appears blue during the day and black at night.
a) Explain the reason for the blue color of the sky.
b) What is the reason why the night sky appears black?
c) Why is darkness experienced in space (outside the Earth's atmosphere)?

28A student sitting at the back of the classroom cannot clearly see the writings on the board.
a) What is the name of this defect?
b) Draw a diagram illustrating this defect.

29.When sunlight passes through a dense forest, the path of light becomes visible.
a) What is this phenomenon?
b) Explain the necessary condition for this phenomenon to occur.

30.How does the eye adjust itself to see nearby objects? Explain the role of the ciliary muscles in this process.

31. In an experiment demonstrating the scattering of light using Sodium Thiosulphate and Hydrochloric Acid:
a) What will be the color of the water in the beaker when viewed from the side?
b) What is the color of the light coming out (transmitted light)?
c) What observation can be made after some time?

32.Newton's color disc appears white when rotated rapidly.
a) What phenomenon of the eye causes this?
b) Define this phenomenon.
c) Give another example of this phenomenon used in daily life.

33..White light is passed through a red filter and allowed to fall on a green book.
a) What is the color of the light emerging from the red filter?
b) What color does the green book reflect?
c) What color will the observer see the book as?

34. You are provided with two prisms.
a) How will you arrange them to get white light back from the spectrum? Draw the diagram.
b) Explain the function of the first and second prisms in this arrangement.

35.. The tail lamps and signal lamps of vehicles use red color.
a) How are the wavelength and scattering of colors related?
b) What is the reason for using red color in signal lamps?

36.. The teacher asked you to plan an experiment to demonstrate the scattering of light.
a) List the materials required for conducting the experiment.
b) Write down the procedure of the experiment

4.MAGNETIC EFFECTS OF ELECTRIC CURRENT

1. The magnetic field around a straight current-carrying conductor consists of
   (a) radial straight lines
   (b) circular lines concentric with the conductor
   (c) parallel straight lines
   (d) irregular closed curves

2. State the right hand thumb rule used to find the direction of magnetic field around a current-carrying conductor.

3. Assertion (A): A solenoid carrying electric current behaves like a bar magnet.
   Reason (R): Magnetic field lines around a solenoid and a bar magnet are similar in shape.
   Choose the correct option:
   (a) Both A and R are true, and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true but R is false.
   (d) A is false but R is true.

4. Choose the correct statements about an electromagnet.
   i) It is made using a soft iron core and a solenoid.
   ii) Its polarity cannot be changed.
   iii) Its magnetic strength can be varied.
   iv) It does not need electric current for magnetism.
   (a) i and ii (b) i and iii (c) ii and iii (d) ii and iv

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5. A straight conductor is placed above a pivoted magnetic needle and connected in series with a battery and a switch.
   a) What happens to the magnetic needle when the switch is turned on?
   b) Name this effect of electric current.

6. A. Draw the magnetic field lines around a bar magnet and mark its poles.

   6. B. Using the right hand thumb rule, explain how you find the direction of magnetic field around a straight current-carrying conductor.

7. A student increased the number of turns of a coil carrying current, keeping the current the same.
   a) What change is observed in the deflection of a nearby magnetic needle?
   b) State the reason for this change.

8. A. Define a solenoid. Write any one similarity between the magnetic field of a solenoid and that of a bar magnet.

   OR

8. B. What is an electromagnet? Write any two factors that increase the strength of an electromagnet.

9. In a current-carrying solenoid, the current at one end is seen to flow in the clockwise direction.
   a) What is the polarity of this end?
   b) Which rule do you use to find this polarity?

10. Complete the table:

Sl. No.	Bar magnet	Current-carrying solenoid
1	Magnetism is permanent.	
2	Polarity cannot be changed.	

11. A conductor AB is placed in a magnetic field such that the direction of current, magnetic field, and force are mutually perpendicular.
   a) Name the rule used to find the direction of force on the conductor.
   b) Write any one application of this rule in an electrical device.

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12. A. A straight current-carrying conductor is placed perpendicular to the plane of a cardboard. Small magnetic compasses are placed around it in a circle.
   a) What happens to the compasses when the current is switched on?
   b) How can you mark the magnetic field lines on the cardboard using these compasses?
   c) From this experiment, write any one inference about the nature of the magnetic field around a straight conductor.

OR

12. B. A student holds a current-carrying conductor with the right hand such that the thumb points in the direction of current.
   a) What does this rule indicate?
   b) How is this rule used to find the direction of magnetic field around the conductor?
   c) State Ampere's swimming rule.

13. Explain how you would show experimentally that the direction of deflection of a magnetic needle placed near a straight current-carrying conductor reverses when the direction of current is reversed.

14. A soft iron core is inserted into a current-carrying solenoid.
   a) What happens to the magnetic field strength?
   b) What is this device called?
   c) Mention two uses of such devices in daily life.

15. A.
   a) State the motor principle.
   b) How is Fleming's left hand rule related to the working of an electric motor?

OR

15.B.

a) State Fleming’s left hand rule.

b) In a motor, why does the armature continue to rotate in the same direction though the current in each arm reverses after every half rotation?

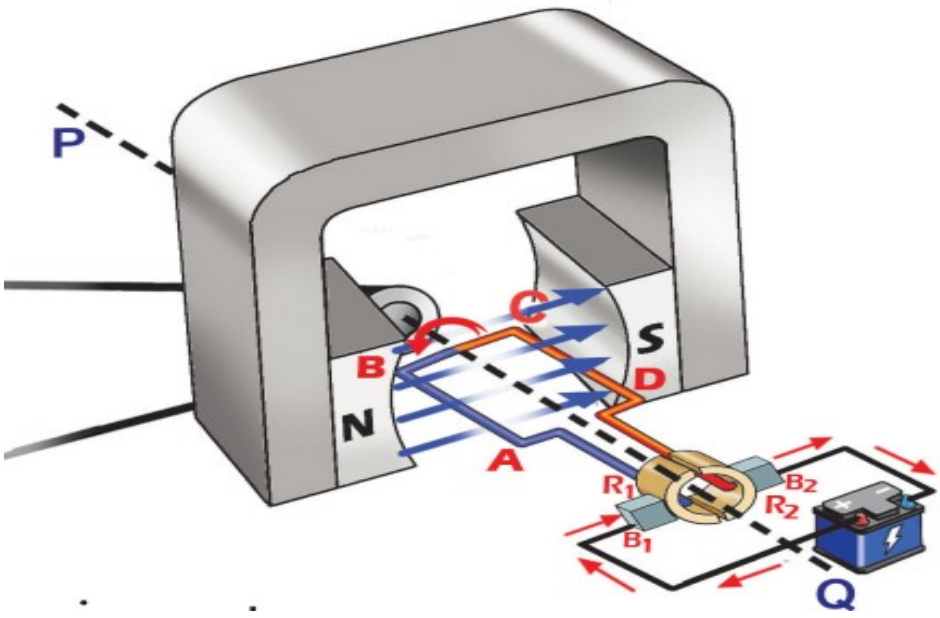
16.The magnetic field around a current-carrying solenoid is compared with that around a bar magnet.

a) Write any two similarities.

b) Write any one difference between them.

17.Explain briefly how a moving coil loudspeaker works based on the motor principle. Mention the main energy conversion taking place in it.

18.A. An electric motor is shown with an armature placed between the poles of a magnet.



a) Name the parts labelled as armature, split ring, and brushes.

b) What is the function of the split ring commutator in this motor?

c) State the energy conversion taking place in this motor.

OR

18.B. An electromagnet is used in a crane to lift heavy loads of scrap iron.

a) Why is a soft iron core used in such an electromagnet instead of a steel core?

b) Write any two advantages of using electromagnets in cranes.

c) State any one precaution to be taken by workers when such strong electromagnets are used in an industry.

Answer Key –

1. (b) circular lines concentric with the conductor
2. In this rule, the conductor is held with the right hand such that the thumb points in the direction of current and the curled fingers give the direction of the magnetic field around the conductor.
3. (a) Both A and R are true, and R is the correct explanation of A.
4. (b) i and iii

5

a) The magnetic needle deflects from its initial position.
b) Magnetic effect of electric current.

6. A.
   Magnetic field lines emerge from the north pole, curve around in closed loops, and enter the south pole; inside the magnet they go from south to north.
6. B.
   By the right hand thumb rule, when the thumb points along the direction of current, the curled fingers show the circular direction of magnetic field lines around the conductor.
7.

a) The deflection of the magnetic needle increases.
b) Increasing the number of turns increases the magnetic field strength of the coil.

8. A.
   A solenoid is a conductor wound in a helical (spiral) form with many turns whose centres lie on a straight line.
   One similarity: The pattern of magnetic field lines around a solenoid and a bar magnet is similar.
8. B.
   An electromagnet is a device that produces a magnetic field using electric current, usually a solenoid with a soft iron core.
   Any two factors:

- Increase the number of turns per unit length.
- Increase the current through the solenoid.
- Use a soft iron core of larger cross-section.

9.

a) That end behaves as a south pole.
b) Right hand rule for solenoids (right hand grip rule).

10.

Sl. No.	Bar magnet	Current-carrying solenoid
1	Magnetism is permanent.	Magnetic strength can be varied.
2	Polarity cannot be changed.	Polarity can be changed by reversing the current. [attached_file

11.

a) Fleming’s left hand rule.
b) Used in electric motors and moving coil loudspeakers (any one).

12.A.

a) The north poles of the compasses deflect tangentially, forming a circular pattern.
b) Mark the direction of the north pole of each needle, remove the compasses, then join these points to draw circular magnetic field lines with arrows.
c) The magnetic field around a straight current-carrying conductor is circular and concentric with the conductor.

12.B.

a) It states that if the conductor is held by the right hand with the thumb along current, the curled fingers show the magnetic field direction around it.
b) Point the thumb in the current direction; the direction in which the fingers curl gives the magnetic field lines.
c) Ampere’s swimming rule: If a person swims in the direction of current facing the magnetic needle, the north pole of the needle deflects towards the swimmer’s left side.

13.

Description should include: setting a straight conductor near a pivoted magnetic needle, observing deflection for a given current direction, then reversing the current and noting that the needle deflects in the opposite sense; conclusion: the direction of deflection depends on the direction of current and reverses with it.

14.

a) The magnetic field strength increases.
b) Electromagnet.
c) Any two uses, e.g. cranes for lifting scrap iron, electric bells, relays, MRI scanners, electric motors.

15.A.

a) Motor principle: A current-carrying conductor placed in a magnetic field and free to move experiences a force causing it to move or deflect.
b) Fleming’s left hand rule gives the direction of the force on the armature conductors in the magnetic field, explaining the direction of rotation of the motor.

15.B.

a) Statement of Fleming’s left hand rule: When the first finger, second finger, and thumb of the left hand are held mutually perpendicular, the first finger gives the field direction, the second finger the current direction, and the thumb the force direction.

b) After each half rotation, the split-ring commutator reverses the current in AB and CD, so that each side coming in front of a given pole always carries current in the same effective direction, making the torque act continuously in the same sense and keeping the armature rotating in one direction.

16.

Any two similarities:

- Both have similar magnetic field line patterns (from north to south outside, closed loops).
- Both have distinct north and south poles.

Any one difference:

- Bar magnet has permanent magnetism; solenoid’s magnetism exists only when current flows and its strength can be varied.

17.

Key points: A voice coil placed in the magnetic field of a permanent magnet carries audio-frequency current; by the motor principle the coil experiences a force and vibrates; the attached diaphragm vibrates, producing sound. Energy conversion: electrical energy to sound energy (via mechanical vibrations).

18.A.

a) Armature: the rotating coil on a soft iron core; split ring: the two semicircular rings connected to armature ends; brushes: graphite or metal blocks pressing on the rings to provide current.
b) The split ring commutator reverses the direction of current through the armature every half rotation, so that torque on the coil is always in the same direction. [attached_file:file:3]
c) Electrical energy → mechanical energy.

18.B.

a) Soft iron is easily magnetised and demagnetised, giving a strong but temporary magnet; steel retains magnetism and is unsuitable where magnetism must be switched on and off.
b) Any two: strength can be varied; can be switched on and off; polarity can be reversed; safer and more controllable lifting of scrap.
c) Example: Workers should keep magnetic objects and personal metal items away from the field region to avoid accidents and unwanted attraction.

CHAPTER 5

ELECTRIC ENERGY CONSUMPTION & CONSERVATION

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1. A 1500 W electric kettle is used for 30 minutes every day. Calculate the electrical energy consumed in kWh for one month (30 days).

2. In a house, five LED bulbs of 9 W each burn for 10 hours daily, and two fans of 60 W each run for 15 hours daily. Find the total energy consumed in a day in 'units'.

3. An electric iron is rated 230 V, 1000 W. If it is used for 2 hours, how many units of electricity does it consume?

4. A 1 HP motor pump (1 HP = 746 W) runs for 2 hours daily to lift water. Calculate the cost of running the motor for 30 days if one unit of electricity costs ₹5.

5. Which consumes more energy: A 100 W TV used for 5 hours or a 1000 W toaster used for 20 minutes?

6. An electric lamp draws a current of 0.5 A when connected to a 220 V supply. Calculate the resistance of the lamp's filament.

7. A bulb is rated 220 V, 100 W. What is its resistance when glowing?

8. Two resistors of 10 Ω and 20 Ω are connected in series to a 6 V battery. Calculate the current flowing through the circuit.

9. Two lamps rated 60 W, 220 V and 100 W, 220 V are connected in parallel to the electric mains supply. What is the total current drawn from the line if the supply voltage is 220 V?

10. Why are household appliances connected in parallel instead of in series? Give two reasons.

11. Calculate the heat generated (in Joules) when a current of 5 A flows through a resistor of 10 Ω for 2 minutes.

12. An electric heater has a resistance of 50 Ω and is connected to a 220 V supply. Calculate the heat produced in 10 minutes in kilojoules (kJ).

13. Why is Nichrome preferred over Copper for making the heating elements of appliances like electric irons and toasters? (Mention two properties).

14. Two heater coils A and B made of the same material have the same length, but the diameter of coil A is twice that of coil B. When connected to the same voltage supply individually, which coil will produce more heat? Explain.

15. A 2 kW immersion water heater is used to heat water. If the voltage supply drops by 10%, by what percentage does the heat production rate decrease?

16. A house has a main fuse rating of 5 A. Can you use a 1.5 kW electric geyser rated at 220 V in this house without blowing the fuse? Justify your answer.

17. Two bulbs are marked 60 W, 220 V and 60 W, 110 V respectively. Write the ratio of their resistance.

18. An electric motor takes 5 A from a 220 V line. The efficiency of the motor is 80%. What is the mechanical power output of the motor?

19. Ramesh replaced all 10 incandescent bulbs (60 W each) in his house with LED bulbs (9 W each). If these bulbs are used for 5 hours daily, calculate the energy saved in units per month (30 days).

20. A heater is designed to operate at 230V. If it is connected to a 110V supply, how will it affect: a) The heat produced per second? b) The resistance of the coil?

6.ELECTROMAGNETIC INDUCTION IN DAILY LIFE

1 MARK

1.Find and fill in the relationship
Transformer : Mutual induction
Generator : .....

2. Which is correct about a step-down transformer?
($V_S > V_P$, $I_S > I_P$, $I_S < I_P$)

3. Write the equation relating the number of turns of a transformer and its voltage?

4. At what voltage does electricity reach our homes?
(II0V , 290V , 230V , 200V)

5. What type of electricity is produced in the armature of a DC generator?
(DC, AC, current at the same voltage, none of these)

2 MARK

6) Make a graphic representation of the electricity generated by AC generator and DC generator.

7) Write one similarity and difference between a nuclear power station and a thermal power station.

8) Write the instructions to be followed in home electrification.

9) Classify the following as those that operate on AC and those that operate on DC.
(Torch, grinder, microwave oven, emergency lamp, calculator)

10) Correct if there is any mistake.
a) The power in the primary and the
secondary of a transformer are equal.
b) A fuse wire is a
device that utilizes the magnetic effect of electricity

3 MARKS

11.Match the following.
DC Generator – Split Rings
AC Generator – Low Melting Point
Fuse Wire – Slip Rings.

12. The ratio of the number of primary turns to the number of secondary turns of a transformer is 2:5.
a) What type of transformer is this?
b) What will be the voltage obtained at the secondary of this transformer if 80v is applied to the primary?
c) What is the ratio between the primary current and the secondary current?

13. Write the energy conversion.
a) Thermal power station -
b) Nuclear power station -
c) Hydroelectric power station -

14. A safety fuse is a device that protects equipment when there is excessive current flowing through a circuit.
a) Write two situations where excessive current flows.
b) In what way is the fuse wire attached to the circuit?

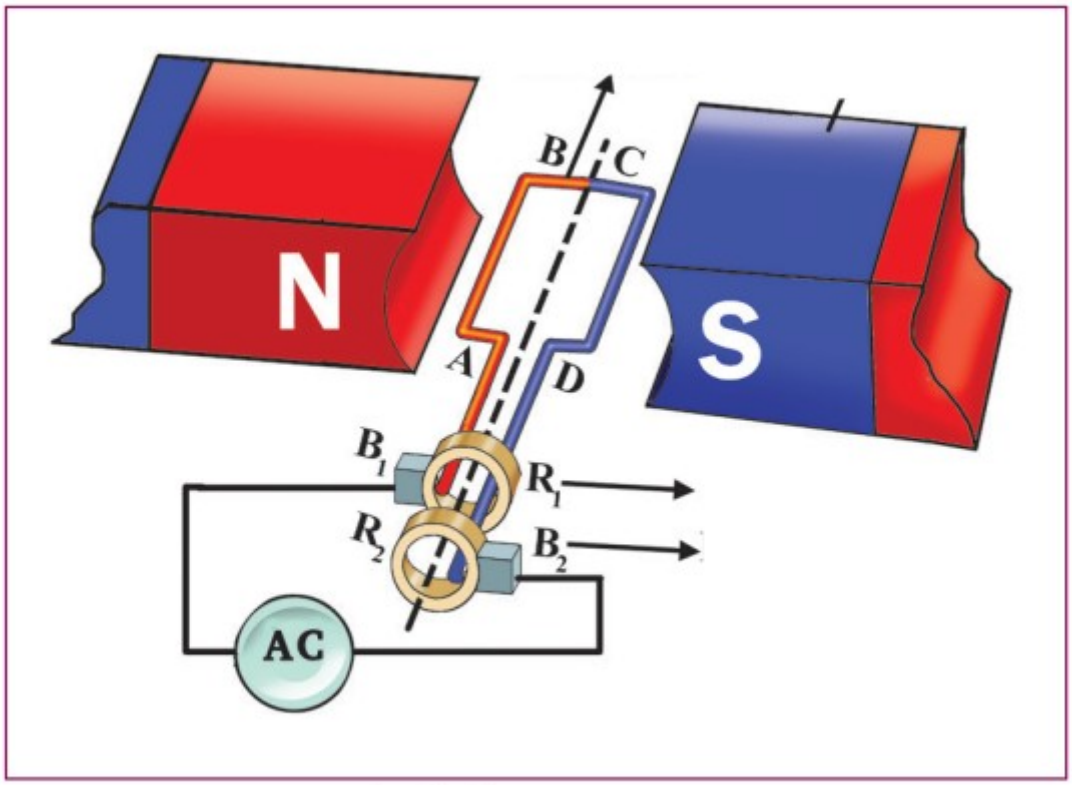
15) Power transmission is the transmission of electricity from power stations to remote locations through wires.
a) At what voltage is electricity generated at a power station?
b) At what stage is a step-up transformer used in power transmission?
c) What are the problems experienced while transmitting electricity to remote locations?

4 MARK

16) Three-pin plugs are used when high-power appliances are used in a circuit.
a) What is each pin in a three-pin plug indicated for?
b) To which part of the appliance is the earth pin connected?
c) How is the safety of a three-pin plug appliance ensured?

17) Write any 4 precautions to avoid electric shock?

18)



AC Generator

a) Which appliance is shown in the diagram?
b) What are R and B in the diagram?
c) Write the working principle of this appliance?
d) Draw a graph of the power generated by this appliance

19) Classify the following statements as step-up transformer and step-down transformer.
a) The number of primary windings is less than the secondary windings
b) Thicker wire windings in the secondary.
c) The current intensity is less in the secondary.
e) Transformer that increases the voltage
f) Transformer that decreases the voltage
g) The number of primary windings
h) The current intensity is more in the secondary

20) Power transmission is the transmission of electricity from power plants to distant places.
a) Which device does the line coming from the electric pole first reach?
b) What is the special feature of the main switch?
c) Suggest a way to ensure safety in domestic electricity supply.

7.Mechanical Advantage in Action

1 MARKS

Question 1

Statement 1: A lever with the fulcrum between the load and effort is called a first order lever.

Statement 2: In a first order lever, the mechanical advantage is always greater than one.

Choose the correct option:

a) Statement 1 and Statement 2 are both correct.
b) Statement 1 and Statement 2 are both incorrect.
c) Statement 1 is correct, but Statement 2 is incorrect.
d) Statement 1 is incorrect, but Statement 2 is correct.

Question 2

When a movable pulley is used to lift a load:

a) The mechanical advantage is always 1.
b) The mechanical advantage is always 2.
c) The mechanical advantage depends on the number of movable pulleys.
d) The mechanical advantage depends on the effort applied.

Question 3

The mechanical advantage of an inclined plane is given by:

a) $MA = \frac{\text{length of inclined plane}}{\text{height}}$
b) $MA = \frac{\text{height}}{\text{length of inclined plane}}$
c) $MA = \frac{\text{weight}}{\text{force applied}}$
d) $MA = \frac{\text{effort arm}}{\text{load arm}}$

Question 4

Statement 1: A screw is essentially an inclined plane wrapped around a cylinder.

Statement 2: The pitch of a screw is the vertical distance between two consecutive threads.

Choose the correct option:

a) Statement 1 and Statement 2 are both correct.
b) Statement 1 and Statement 2 are both incorrect.
c) Statement 1 is correct, but Statement 2 is incorrect.
d) Statement 1 is incorrect, but Statement 2 is correct.

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2 MARKS

Question 5

Define mechanical advantage. A nail puller exerts a force of 50 N to pull a nail. If the mechanical advantage of the nail puller is 4, what is the force exerted by the nail?

Question 6

With the help of a neat diagram, distinguish between a first order lever and a second order lever. Give one example each.

Question 7

Explain how gears in a bicycle help to change the speed and power of the vehicle while climbing uphill.

Question 8

A meter scale is suspended at its midpoint. A load of 60 gwt is suspended at a distance of 20 cm from the fulcrum on one side. At what distance should a load of 40 gwt be suspended on the other side to balance the scale? (Use the principle of Levers)

Question 9

Explain why a longer inclined plane makes it easier to move a load to a certain height compared to a shorter inclined plane. Is there a gain in work done?

Question 10

How is a wheel and axle similar to a lever? Explain with the help of a diagram showing how train bogies were lifted from a lake using a wheel and axle.

Question 11

A load of 800 N is to be lifted to a height of 4 m using an inclined plane. Calculate:
a) The work done in lifting the load directly.

b) If a force of 200 N is applied along an inclined plane to lift the load by 4 m, what is the length of the inclined plane?

Question 12

Compare fixed pulley and movable pulley in terms of mechanical advantage and their practical applications. Which one would you use to lift a heavy load and why?

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3 MARKS

Question 13

Explain the principle of a lever with the help of the lever principle equation. An object weighing 1200 N is placed at a distance of 0.5 m from the fulcrum on one side of a first order lever. If the effort arm is 2 m, what effort must be applied on the other side to lift the object?

Question 14

State and explain the three types of levers with one example each from daily life. Which type of lever is used in a pair of scissors and why?

Question 15

A system of pulleys consists of 3 movable pulleys arranged in series (first type system). A load of 2400 N is to be lifted using this pulley system. Calculate:
a) The mechanical advantage of the system.
b) The effort required to lift the load.
c) If the effort is applied through a distance of 12 m, how much will the load rise?

Question 16

Define centre of gravity. Explain with a diagram how an object can be balanced on a knife edge at its centre of gravity. What is the significance of centre of gravity in the functioning of a lever?

Question 17

Describe the structure and working of a screw jack. Explain why a screw is considered a simple machine and how its mechanical advantage can be calculated experimentally.

Question 18

Differentiate between a wedge and a screw. How are they related to inclined planes? Give practical examples of each from your daily life.

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4 MARKS

Question 19

A log weighing 6000 kg is to be lifted through a height of 3 m onto a truck using an inclined plane. A force of 2000 N is applied. Calculate:
a) The length of the inclined plane required.
b) The mechanical advantage.
c) The work done by the effort applied.
d) Is there a gain in work? Justify your answer. (Consider g = 10 m/s^2)

ANSWER KEY

SECTION A (Answers)

Q. No.	Answer
1	c) Statement 1 is correct, but Statement 2 is incorrect.
2	c) The mechanical advantage depends on the number of movable pulleys.
3	b) $MA = \frac{\text{length of inclined plane}}{\text{height}}$
4	a) Statement 1 and Statement 2 are both correct.

SECTION B

Q.5: Mechanical advantage is the ratio of load to effort. It indicates how many times the machine multiplies the applied effort.

Given: Effort = 50 N, MA = 4
Force exerted by nail = MA × Effort = 4 × 50 = **200 N**

Q.8: Using principle of moments: $Load_1 \times Distance_1 = Load_2 \times Distance_2$
$60 \times 20 = 40 \times d$
$d = 1200/40 =$ **30 cm**

Q.11:
a) Work = Force × Height = 800 × 4 = **3200 J**
b) Length = (Effort × Distance) / Load = (200 × 4) / 800 = **2.67 m** (or approximately **2.67 m**)

SECTION C

Q.13: Lever principle: Load × Load arm = Effort × Effort arm
1200 × 0.5 = Effort × 2
Effort = 600/2 = **300 N**

Q.15:
a) For 3 movable pulleys: $MA = 2^3 =$ **8**
b) Effort = Load/MA = 2400/8 = **300 N**
c) Load rises = Effort distance/2 = 12/2 = **6 m**

SECTION D

Q.19:

a) Using Work principle: Effort × Distance = Load × Height
2000 × l = 6000 × 10 × 3
l = 180,000/2000 = **90 m**

b) MA = l/h= 90/3 = **30**

c) Work by effort = 2000 × 90 = **1,80,000 J**

d) **No gain in work.** Work done on load = 6000 × 10 × 3 = 1,80,000 J. This equals work by effort, confirming no gain in work by simple machines.
